Week 3 complete solutions

Problem-sheet solutions

Each solution states the intended method and conditions, shows the marked reasoning and gives specific common mistakes. Award proportionate method credit after a purely arithmetic slip when the setup remains valid.

Question 1 — membership in a span

`W03-Q01-PA (M5)` **Method and conditions.** Membership in the span requires real numbers a, b such that $a\mathbf{v}_1 + b\mathbf{v}_2 = \mathbf{w}$. The three coordinate equations must be mutually consistent.

The coefficient equation is

$$a\begin{bmatrix}1\\0\\1\end{bmatrix} + b\begin{bmatrix}0\\1\\1\end{bmatrix} = \begin{bmatrix}2\\-1\\1\end{bmatrix}.$$

The first two coordinates give $a = 2$ and $b = -1$. The third coordinate requires $a + b = 1$, which these values satisfy. Direct verification gives

$$2\mathbf{v}_1 - \mathbf{v}_2 = \begin{bmatrix}2\\0\\2\end{bmatrix} - \begin{bmatrix}0\\1\\1\end{bmatrix} = \begin{bmatrix}2\\-1\\1\end{bmatrix} = \mathbf{w}.$$

Final result. $\mathbf{w} \in \operatorname{span}\{\mathbf{v}_1, \mathbf{v}_2\}$, with coefficients $a = 2$ and $b = -1$.

Marks (5). Correct coefficient equation, 1; first two coordinate conditions, 1; coefficients, 1; consistency check, 1; verified conclusion, 1.

Common mistakes. Comparing only two coordinates; treating the coefficients as vector entries without checking the linear combination; omitting the final consistency check.

Question 2 — independence, basis and coordinates

`W03-Q02-PA (M5)` **Method and conditions.** A non-zero coefficient relation establishes linear dependence. The dimension of the span is the number of vectors in a basis for that span.

The vectors satisfy

$$\mathbf{u}_3 = \mathbf{u}_1 + \mathbf{u}_2,$$

or equivalently,

$$\mathbf{u}_1 + \mathbf{u}_2 - \mathbf{u}_3 = \mathbf{0}.$$

The coefficients $(1, 1, -1)$ are not all zero, so the set is linearly dependent.

The two vectors $\mathbf{u}_1$ and $\mathbf{u}_2$ are not scalar multiples and are therefore independent. Since $\mathbf{u}_3$ is already in their span,

$$\operatorname{span}\{\mathbf{u}_1, \mathbf{u}_2, \mathbf{u}_3\} = \operatorname{span}\{\mathbf{u}_1, \mathbf{u}_2\}.$$

Final result. The three-vector set is linearly dependent and its span has dimension 2.

Marks (5). Valid dependence relation, 2; correct dependence conclusion, 1; independence of the remaining pair, 1; dimension conclusion, 1.

Common mistakes. Calling the set independent because none of the vectors is zero; giving a relation whose coefficients are all zero; reporting three as the dimension merely because three vectors were listed.

`W03-Q02-PB (M5)` **Method and conditions.** Three independent vectors in $\mathbb{R}^3$ form a basis. A non-zero determinant tests independence. Coordinates are the coefficients in the ordered basis.

Put the basis candidates into the columns of

$$\mathbf{P} = \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 1 \\ 1 & 1 & 0 \end{bmatrix}.$$

Its determinant is

$$\det(\mathbf{P}) = 1(0-1) + 1(0-1) = -2 \neq 0.$$

Thus the three columns are independent and form a basis of $\mathbb{R}^3$.

For the coordinates, solve

$$a\mathbf{u}_1 + b\mathbf{u}_2 + c\mathbf{u}_4 = \begin{bmatrix} 3 \\ 1 \\ 2 \end{bmatrix}.$$

The coordinate equations are

$$a + c = 3, \qquad b + c = 1, \qquad a + b = 2.$$

Adding the first two equations and subtracting the third gives $2c = 2$, so $c = 1$. Then $a = 2$ and $b = 0$.

Final result and interpretation.

$$[\mathbf{x}]_{\mathcal{B}} = \begin{bmatrix} 2 \\ 0 \\ 1 \end{bmatrix}.$$

The representation is unique because a second coefficient triple would give a non-zero linear dependence relation among the basis vectors after subtraction.

Marks (6). Correct determinant or equivalent independence test, 2; basis conclusion, 1; coefficient equations, 1; coordinates, 1; uniqueness explanation, 1.

Common mistakes. Putting the candidate basis vectors into rows without adjusting the coordinate equation; treating spanning alone as sufficient for a basis; changing the coordinate order; asserting uniqueness without using independence.

Question 3 — eigenvalues, eigenvectors and a diagonalisation

`W03-Q03-PA (M5)` **Method and conditions.** Eigenvalues are roots of $\det(\mathbf{A} - \lambda\mathbf{I}_2) = 0$.

$$\begin{aligned}\det(\mathbf{A}-\lambda\mathbf{I}_2) &= \begin{vmatrix} 4-\lambda & 1 \\ 2 & 3-\lambda \end{vmatrix} \\ &= (4-\lambda)(3-\lambda)-2 \\ &= \lambda^2 - 7\lambda + 10 \\ &= (\lambda-5)(\lambda-2).\end{aligned}$$

Final result. The characteristic polynomial is $\lambda^2 - 7\lambda + 10$, and the eigenvalues are 5 and 2.

Marks (5). Correct characteristic determinant, 1; expansion, 2; factorisation, 1; both eigenvalues, 1.

Common mistakes. Using $\det(\mathbf{A})-\lambda$ as the characteristic polynomial; losing the off-diagonal product; reporting diagonal entries as eigenvalues without calculation.

`W03-Q03-PB (M5)` **Method and conditions.** For each eigenvalue, solve the homogeneous system $(\mathbf{A}-\lambda\mathbf{I}_2)\mathbf{v} = \mathbf{0}$. A diagonalisation requires an invertible matrix of eigenvectors.

For $\lambda = 5$,

$$\begin{bmatrix} -1 & 1 \\ 2 & -2 \end{bmatrix}\begin{bmatrix} v_1 \\ v_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix},$$

so $v_2 = v_1$. Choose $\mathbf{v}_1 = [1, 1]^\mathsf{T}$.

For $\lambda = 2$,

$$\begin{bmatrix} 2 & 1 \\ 2 & 1 \end{bmatrix}\begin{bmatrix} v_1 \\ v_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix},$$

so $v_2 = -2v_1$. Choose $\mathbf{v}_2 = [1, -2]^\mathsf{T}$.

Place these vectors and their eigenvalues in the same order:

$$\mathbf{P} = \begin{bmatrix} 1 & 1 \\ 1 & -2 \end{bmatrix}, \qquad \mathbf{D} = \begin{bmatrix} 5 & 0 \\ 0 & 2 \end{bmatrix}.$$

Then

$$\mathbf{AP} = \begin{bmatrix} 5 & 2 \\ 5 & -4 \end{bmatrix} = \mathbf{PD}.$$

Also, $\det(\mathbf{P}) = -3 \neq 0$, so $\mathbf{P}$ is invertible, with

$$\mathbf{P}^{-1} = \begin{bmatrix} 2/3 & 1/3 \\ 1/3 & -1/3 \end{bmatrix}.$$

Multiplying $\mathbf{AP} = \mathbf{PD}$ on the right by $\mathbf{P}^{-1}$ gives the diagonalisation.

Final result.

$$\mathbf{A} = \mathbf{PDP}^{-1}$$

for the matrices displayed above.

Marks (7). Eigenvector for $\lambda = 5$, 1; eigenvector for $\lambda = 2$, 1; consistent $\mathbf{P}$ and $\mathbf{D}$, 2; verification $\mathbf{AP} = \mathbf{PD}$, 1; invertibility check, 1; diagonalisation conclusion, 1.

Common mistakes. Allowing the zero vector as an eigenvector; pairing an eigenvector with the wrong diagonal entry; assuming $\mathbf{P}$ is invertible without checking independence; reversing the order in $\mathbf{PDP}^{-1}$.

Question 4 — diagonalisation and eigenbasis coordinates

`W03-Q04-PA (M5)` **Method and conditions.** Find two independent real eigenvectors. The ordered columns of $\mathbf{P}$ define the eigenbasis, so $[\mathbf{u}]_{\mathcal{B}} = \mathbf{P}^{-1}\mathbf{u}$ and $\mathbf{Bu} = \mathbf{PD}[\mathbf{u}]_{\mathcal{B}}$.

The eigenvalues are 3 and 2. For $\lambda = 3$, an eigenvector is $[1, 0]^{\mathsf{T}}$. For $\lambda = 2$, the equation $v_1 + v_2 = 0$ gives the eigenvector $[-1, 1]^{\mathsf{T}}$.

Use

$$\mathbf{P} = \begin{bmatrix} 1 & -1 \\ 0 & 1 \end{bmatrix}, \qquad \mathbf{D} = \begin{bmatrix} 3 & 0 \\ 0 & 2 \end{bmatrix}, \qquad \mathbf{P}^{-1} = \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}.$$

Since $\det(\mathbf{P}) = 1$, this is a valid diagonalisation. Let $\mathbf{u} = [5, 2]^{\mathsf{T}}$. Its coordinates in the ordered eigenbasis are

$$[\mathbf{u}]_{\mathcal{B}} = \mathbf{P}^{-1}\mathbf{u} = \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} 5 \\ 2 \end{bmatrix} = \begin{bmatrix} 7 \\ 2 \end{bmatrix}.$$

Therefore

$$\mathbf{Bu} = \mathbf{PD}[\mathbf{u}]_{\mathcal{B}} = \begin{bmatrix} 1 & -1 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} 21 \\ 4 \end{bmatrix} = \begin{bmatrix} 17 \\ 4 \end{bmatrix}.$$

The direct check agrees:

$$\begin{bmatrix} 3 & 1 \\ 0 & 2 \end{bmatrix} \begin{bmatrix} 5 \\ 2 \end{bmatrix} = \begin{bmatrix} 17 \\ 4 \end{bmatrix}.$$

Final result. The ordered eigenbasis is $\{[1, 0]^{\mathsf{T}}, [-1, 1]^{\mathsf{T}}\}$, $[\mathbf{u}]_{\mathcal{B}} = [7, 2]^{\mathsf{T}}$, and $\mathbf{Bu} = [17, 4]^{\mathsf{T}}$.

Marks (10). Eigenvalues, 1; two correctly paired eigenvectors, 2; valid $\mathbf{P}$, $\mathbf{D}$ and inverse, 2; eigenbasis coordinates, 2; $\mathbf{PD}[\mathbf{u}]_{\mathcal{B}}$, 1; correct vector, 1; direct or order/validity check, 1.

Common mistakes. Treating the diagonal entries alone as a diagonalisation; mismatching eigenvector and eigenvalue order; using $\mathbf{Pu}$ rather than $\mathbf{P}^{-1}\mathbf{u}$ for eigenbasis coordinates; applying $\mathbf{D}$ to standard coordinates.

Question 5 — a PMF, its CDF and moments

`W03-Q05-PA (S3)` **Method and conditions.** A PMF is non-negative on its support and sums to one. The CDF accumulates masses at and below its argument.

Normalisation gives

$$1 = \sum_{x=0}^{3} c(x + 1) = c(1 + 2 + 3 + 4) = 10c,$$

so $c = 1/10$. The CDF is

$$F_X(x) = \begin{cases} 0, & x < 0, \\ 1/10, & 0 \le x < 1, \\ 3/10, & 1 \le x < 2, \\ 6/10, & 2 \le x < 3, \\ 1, & x \ge 3. \end{cases}$$

The event $1 < X \le 3$ consists of $X = 2$ and $X = 3$, so

$$\mathbb{P}(1 < X \le 3) = \frac{3}{10} + \frac{4}{10} = \frac{7}{10}.$$

Final result. $c = 1/10$, with the CDF and probability shown above.

Marks (5). Correct normalisation and c, 1; complete piecewise CDF, 3; event probability, 1.

Common mistakes. Normalising the support values rather than the PMF values; omitting the right-continuous endpoint convention; treating a discrete interval probability as a length; excluding the mass at $X = 3$.

`W03-Q05-PB (S3, S4)` **Method and conditions.** Calculate the first two raw moments from the PMF. For a linear transformation, expectation is linear and variance squares the multiplicative coefficient.

$$\mathbb{E}[X] = \frac{0(1) + 1(2) + 2(3) + 3(4)}{10} = \frac{20}{10} = 2,$$

and

$$\mathbb{E}[X^2] = \frac{0^2(1) + 1^2(2) + 2^2(3) + 3^2(4)}{10} = \frac{50}{10} = 5.$$

Therefore,

$$\operatorname{Var}(X) = 5 - 2^2 = 1.$$

For $Y = 3 - 2X$,

$$\mathbb{E}[Y] = 3 - 2\mathbb{E}[X] = -1$$

and

$$\operatorname{Var}(Y) = (-2)^2 \operatorname{Var}(X) = 4.$$

Final result and interpretation.

$$\mathbb{E}[X] = 2, \quad \operatorname{Var}(X) = 1, \quad \mathbb{E}[Y] = -1, \quad \operatorname{Var}(Y) = 4.$$

The additive constant shifts every value and therefore shifts the mean. It does not change deviations from the mean, so it does not change variance.

Marks (6). Correct mean of X, 1; second raw moment, 1; variance of X, 1; transformed mean, 1; transformed variance, 1; explanation, 1.

Common mistakes. Using $\mathrm{Var}(X) = \mathbb{E}[X^2] - \mathbb{E}[X]$; failing to square the coefficient -2; adding 3 to the variance; calculating moments with CDF values instead of PMF values.

Question 6 — required discrete distributions

`W03-Q06-PA (S3)` **Method and conditions.** The binomial model counts successes in a fixed number of independent Bernoulli trials. Under the course convention, the geometric variable counts trials through the first success.

The supports are

$$K \in \{0, 1, 2, 3, 4\}, \qquad G \in \{1, 2, ...\}.$$

For the binomial variable,

$$\mathbb{P}(K = 2) = \binom{4}{2} \left(\frac{1}{2}\right)^2 \left(\frac{1}{2}\right)^2 = \frac{6}{16} = \frac{3}{8}.$$

Also,

$$\mathbb{P}(K \geq 1) = 1 - \mathbb{P}(K = 0) = 1 - \left(\frac{1}{2}\right)^4 = \frac{15}{16}.$$

For $G > 3$, the first three trials must all fail. Thus

$$\mathbb{P}(G > 3) = \left(1 - \frac{1}{4}\right)^3 = \left(\frac{3}{4}\right)^3 = \frac{27}{64}.$$

Final result. The supports and three probabilities are as displayed above.

Marks (5). Two correct supports, 1; binomial point probability, 1; binomial complement probability, 1; geometric event representation, 1; geometric probability, 1.

Common mistakes. Using ordered arrangements instead of the binomial coefficient; omitting $K = 4$ from the support; using a geometric support beginning at zero; treating $G > 3$ as two failures rather than three.

`W03-Q06-PB (S3, S4)` **Method and conditions.** Use the Poisson PMF on $\{0, 1, 2, ...\}$, then use the stated parameterisations for the means and variances.

$$\begin{aligned} \mathbb{P}(N \leq 1) &= \mathbb{P}(N = 0) + \mathbb{P}(N = 1) \\ &= e^{-2}\frac{2^0}{0!} + e^{-2}\frac{2^1}{1!} \\ &= 3e^{-2}. \end{aligned}$$

For $N \sim \text{Poisson}(2)$,

$$\mathbb{E}[N] = 2, \qquad \mathrm{Var}(N) = 2.$$

For $K \sim \text{Binomial}(4, 1/2)$,

$$\mathbb{E}[K] = 4\left(\frac{1}{2}\right) = 2, \qquad \mathrm{Var}(K) = 4\left(\frac{1}{2}\right)\left(\frac{1}{2}\right) = 1.$$

Final result and interpretation. The variables have the same mean, 2, but variances 2 and 1. The mean does not determine a full distribution: it records only its centre, whereas variance records spread.

Marks (7). Correct Poisson support or PMF, 1; probability calculation, 2; Poisson moments, 1; binomial moments, 2; comparison and explanation, 1.

Common mistakes. Omitting the factorial in the Poisson PMF; treating the Poisson parameter as a success probability; using np as the binomial variance; claiming that equal means imply equal distributions.

Question 7 — required continuous distributions

`W03-Q07-PA (S3)` **Method and conditions.** A uniform PDF is constant on its closed support and integrates to one. The CDF is the integrated density.

The support is $[-1, 3]$, and

$$f_U(u) = \begin{cases} 1/4, & -1 \le u \le 3, \\ 0, & \text{otherwise.} \end{cases}$$

The CDF is

$$F_U(u) = \begin{cases} 0, & u < -1, \\ (u+1)/4, & -1 \le u < 3, \\ 1, & u \ge 3. \end{cases}$$

Therefore,

$$\mathbb{P}(0 < U \le 2) = \int_0^2 \frac{1}{4}\,\mathrm{d}u = \frac{1}{2}.$$

The inclusion or exclusion of endpoints does not change a continuous probability.

Final result. The support, PDF, CDF and probability are given above.

Marks (5). Correct support and PDF, 1; complete piecewise CDF, 3; interval probability, 1.

Common mistakes. Using density $1/(a+b)$ instead of $1/(b-a)$; omitting the zero-density region; failing to shift by the lower endpoint in the CDF; treating a density value as the interval probability.

`W03-Q07-PB (S3, S4)` **Method and conditions.** The parameter 2 is the exponential rate, so the support is $[0, \infty)$ and the density is $2e^{-2t}$. Use the survival integral and the linear-transformation rules.

$$f_T(t) = \begin{cases} 2e^{-2t}, & t \ge 0, \\ 0, & t < 0. \end{cases}$$

The tail probability is

$$\mathbb{P}(T > 1) = \int_1^\infty 2e^{-2t}\,\mathrm{d}t = e^{-2}.$$

Under the rate parameterisation,

$$\mathbb{E}[T] = \frac{1}{2}, \qquad \mathrm{Var}(T) = \frac{1}{4}.$$

Thus, for $Y = 4T - 1$,

$$\mathbb{E}[Y] = 4\left(\frac{1}{2}\right) - 1 = 1$$

and

$$\mathrm{Var}(Y) = 4^2\left(\frac{1}{4}\right) = 4.$$

Final result.

$$\mathbb{P}(T > 1) = e^{-2}, \quad \mathbb{E}[T] = \frac{1}{2}, \quad \mathrm{Var}(T) = \frac{1}{4}, \quad \mathbb{E}[Y] = 1, \quad \mathrm{Var}(Y) = 4.$$

Marks (7). Support and PDF, 1; tail setup and value, 2; mean and variance of T, 2; transformed mean, 1; transformed variance, 1.

Common mistakes. Treating 2 as the exponential mean; allowing negative support; using $1 - e^{-2}$ for the upper tail; multiplying variance by 4 rather than 4^2.

Question 8 — covariance, correlation and linear transformations

`W03-Q08-PA (S4)` **Method and conditions.** Finite second moments make the variances and covariance well defined. Standardise covariance to obtain correlation, then use bilinearity of covariance and the variance-of-a-linear-combination formula.

The standard deviations are 2 and 3, so

$$\mathrm{Corr}(X, Y) = \frac{-3}{2 \cdot 3} = -\frac{1}{2}.$$

Linearity of expectation gives

$$\mathbb{E}[U] = 2\mathbb{E}[X] - \mathbb{E}[Y] = 2(1) - 2 = 0.$$

For the variances,

$$\begin{aligned} \mathrm{Var}(U) &= 4\,\mathrm{Var}(X) + \mathrm{Var}(Y) - 4\,\mathrm{Cov}(X, Y) \\ &= 4(4) + 9 - 4(-3) \\ &= 37, \end{aligned}$$

and

$$\begin{aligned} \mathrm{Var}(V) &= \mathrm{Var}(X) + \mathrm{Var}(Y) + 2\,\mathrm{Cov}(X, Y) \\ &= 4 + 9 + 2(-3) \\ &= 7. \end{aligned}$$

Finally,

$$\begin{aligned}\operatorname{Cov}(U,V) &= \operatorname{Cov}(2X-Y, X+Y)\\ &= 2\operatorname{Var}(X) + 2\operatorname{Cov}(X,Y) - \operatorname{Cov}(Y,X) - \operatorname{Var}(Y)\\ &= 8 - 6 + 3 - 9\\ &= -4.\end{aligned}$$

Final result.

$$\operatorname{Corr}(X,Y) = -\frac{1}{2}, \quad \mathbb{E}[U] = 0, \quad \operatorname{Var}(U) = 37, \quad \operatorname{Var}(V) = 7, \quad \operatorname{Cov}(U,V) = -4.$$

Marks (8). Correlation, 1; transformed expectation, 1; valid variance formulae, 2; two variances, 2; covariance expansion and value, 2.

Common mistakes. Dividing covariance by the product of variances rather than standard deviations; losing the sign of the covariance term in $\operatorname{Var}(2X-Y)$; assuming covariance is unchanged by scaling; omitting cross-covariance terms.

Question 9 — indicator variables and sampling without replacement

`W03-Q09-PA (S4)` **Method and conditions.** Each indicator is Bernoulli. Use symmetry for its marginal probability, the multiplication rule without replacement for the joint probability, and the covariance term in the variance of a sum.

For $j = 1, 2$,

$$\mathbb{P}(I_j = 1) = \frac{1}{2},$$

so

$$\mathbb{E}[I_j] = \frac{1}{2}, \qquad \operatorname{Var}(I_j) = \frac{1}{2}\left(1 - \frac{1}{2}\right) = \frac{1}{4}.$$

Also,

$$\mathbb{E}[I_1 I_2] = \mathbb{P}(I_1 = 1, I_2 = 1) = \frac{1}{2}\frac{1}{3} = \frac{1}{6}.$$

Therefore,

$$\operatorname{Cov}(I_1, I_2) = \frac{1}{6} - \frac{1}{2}\frac{1}{2} = -\frac{1}{12}.$$

Linearity of expectation gives

$$\mathbb{E}[R] = \mathbb{E}[I_1] + \mathbb{E}[I_2] = 1.$$

The variance is

$$\begin{aligned}\operatorname{Var}(R) &= \operatorname{Var}(I_1) + \operatorname{Var}(I_2) + 2\operatorname{Cov}(I_1, I_2)\\ &= \frac{1}{4} + \frac{1}{4} - \frac{1}{6}\\ &= \frac{1}{3}.\end{aligned}$$

Finally,

$$\mathrm{Corr}(I_1, I_2) = \frac{-1/12}{(1/2)(1/2)} = -\frac{1}{3}.$$

Final result and interpretation.

$$\mathbb{E}[R] = 1, \qquad \mathrm{Var}(R) = \frac{1}{3}, \qquad \mathrm{Corr}(I_1, I_2) = -\frac{1}{3}.$$

The correlation is negative because drawing a red card first reduces the remaining proportion of red cards on the second draw.

Marks (8). Indicator means and variances, 2; joint probability, 1; covariance, 1; expectation of R, 1; variance of R, 1; correlation, 1; contextual interpretation, 1.

Common mistakes. Treating the two draws as independent; using $(1/2)^2$ for the joint probability; omitting covariance from the variance of the sum; interpreting negative correlation as an impossible second red draw.

Question 10 — earlier-outcome retrieval

`W03-Q10-PA` **(M4 retrieval) Method and conditions.** Use row dependence to identify the ranks, compare their common value with the number of unknowns and then parameterise the solution set.

The second equation is twice the first. The first and third coefficient rows are independent, so

$$\mathrm{rank}(\mathbf{A}) = 2.$$

The right-hand side of the second equation is also twice that of the first, and no contradiction appears. Hence

$$\mathrm{rank}([\mathbf{A} \mid \mathbf{b}]) = 2.$$

Since the common rank is less than the three unknowns, the system has multiple solutions. The third equation gives $x = y$. Let $x = y = t \in \mathbb{R}$. The first equation then gives $z = 2 - 2t$.

Final result.

$$\begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} t \\ t \\ 2 - 2t \end{bmatrix}, \qquad t \in \mathbb{R}.$$

Marks (3). Two correct ranks and classification, 1; valid free parameter and complete solution, 1; substitution or equivalent consistency check, 1.

Common mistakes. Counting the repeated equation as a new pivot; calling the system unique because it is consistent; omitting the real parameter domain; leaving one variable unspecified.

`W03-Q10-PB` **(S2 retrieval) Method and conditions.** The events A, A^c form a partition, so total probability gives the denominator for Bayes' rule.

$$\begin{aligned}\mathbb{P}(B) &= \mathbb{P}(B \mid A)\mathbb{P}(A) + \mathbb{P}(B \mid A^c)\mathbb{P}(A^c) \\ &= \frac{3}{4}\frac{1}{3} + \frac{1}{4}\frac{2}{3} \\ &= \frac{1}{4} + \frac{1}{6} \\ &= \frac{5}{12}.\end{aligned}$$

Since $\mathbb{P}(B) = 5/12 > 0$, the posterior is defined. Bayes' rule gives

$$\mathbb{P}(A \mid B) = \frac{\mathbb{P}(B \mid A)\mathbb{P}(A)}{\mathbb{P}(B)} = \frac{1/4}{5/12} = \frac{3}{5}.$$

Final result. $\mathbb{P}(A \mid B) = 3/5$.

Marks (4). Complement prior and total-probability setup, 1; correct $\mathbb{P}(B)$, 1; Bayes setup, 1; positivity condition and posterior, 1.

Common mistakes. Reversing the supplied conditional probability; omitting the A^c branch; using $\mathbb{P}(A)$ as the denominator; failing to check that the conditioning event has positive probability.

`W03-Q10-PC (M2 retrieval)` **Method and conditions.** State the real-log domain before applying the quotient rule, then retain only admissible algebraic values.

The domain requires

$$x - 1 > 0 \quad \text{and} \quad x > 0,$$

so $x > 1$. On this domain,

$$\log\left(\frac{x-1}{x}\right) = \log\left(\frac{1}{3}\right).$$

The natural logarithm is one-to-one on the positive reals, so

$$\frac{x-1}{x} = \frac{1}{3}.$$

Thus

$$3x - 3 = x, \qquad 2x = 3, \qquad x = \frac{3}{2}.$$

This value satisfies $x > 1$. Substitution gives

$$\log\left(\frac{1}{2}\right) - \log\left(\frac{3}{2}\right) = \log\left(\frac{1/2}{3/2}\right) = \log\left(\frac{1}{3}\right).$$

Final result. The unique real solution is $x = 3/2$.

Marks (4). Complete domain, 1; valid logarithm quotient and injectivity, 1; algebraic solution, 1; admissibility and original-equation check, 1.

Common mistakes. Combining logarithms before enforcing positivity; writing $\log(x-1) - \log x = \log(-1)$; cross-multiplying incorrectly; omitting the check against the original logarithms.

Self-test solutions and error check

Solutions and error check

Solution to `W03-ST01-PA (M5)` Method and conditions. Solve the characteristic equation, calculate one non-zero vector in each eigenspace and check independence of the chosen eigenvectors.

Because $\mathbf{C}$ is triangular, its eigenvalues are 2 and 3. For $\lambda = 2$, the equation

$$(\mathbf{C} - 2\mathbf{I}_2)\mathbf{v} = \begin{bmatrix} 0 & 1 \\ 0 & 1 \end{bmatrix} \mathbf{v} = \mathbf{0}$$

gives the eigenvector $[1, 0]^\mathsf{T}$. For $\lambda = 3$,

$$(\mathbf{C} - 3\mathbf{I}_2)\mathbf{v} = \begin{bmatrix} -1 & 1 \\ 0 & 0 \end{bmatrix} \mathbf{v} = \mathbf{0},$$

so one eigenvector is $[1, 1]^\mathsf{T}$.

Thus one valid choice is

$$\mathbf{P} = \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}, \qquad \mathbf{D} = \begin{bmatrix} 2 & 0 \\ 0 & 3 \end{bmatrix}.$$

Since $\det(\mathbf{P}) = 1 \neq 0$, the eigenvectors are independent and $\mathbf{C} = \mathbf{PDP}^{-1}$.

Marks (5). Eigenvalues, 1; two eigenvectors, 2; consistent $\mathbf{P}$ and $\mathbf{D}$, 1; invertibility and diagonalisation conclusion, 1.

Common mistakes. Pairing an eigenvector with the wrong diagonal entry; using the zero vector; assuming a triangular matrix is already diagonal; omitting the independence check.

Solution to `W03-ST02-PA (S3)` Method and conditions. Under $\mathcal{N}(\mu, \sigma^2)$, the second parameter is the variance. Standardise using $Z = (X - \mu)/\sigma$.

The support is $\mathbb{R}$, the variance is 4, and the standard deviation is 2. Therefore,

$$\mathbb{P}(8 \leq X \leq 12) = \mathbb{P}\left(-1 \leq \frac{X - 10}{2} \leq 1\right) = \Phi(1) - \Phi(-1).$$

By symmetry,

$$\Phi(-1) = 1 - \Phi(1) = 0.1587.$$

Hence

$$\mathbb{P}(8 \leq X \leq 12) = 0.8413 - 0.1587 = 0.6826.$$

Marks (5). Support and standard deviation, 1; correct standardisation, 2; symmetry step, 1; final probability, 1.

Common mistakes. Treating 4 as the standard deviation; omitting the mean when standardising; adding the two CDF values; using a support restricted to non-negative values.

Solution to `W03-ST03-PA (S4)` Method and conditions. Use linearity of expectation, the variance rule for a linear transformation and $\mathbb{E}[Y^2] = \mathrm{Var}(Y) + \mathbb{E}[Y]^2$.

$$\mathbb{E}[Y] = 5 - 2\mathbb{E}[X] = 5 - 4 = 1.$$

Also,

$$\mathrm{Var}(Y) = (-2)^2 \,\mathrm{Var}(X) = 4(3) = 12.$$

Therefore,

$$\mathbb{E}[Y^2] = \mathrm{Var}(Y) + \mathbb{E}[Y]^2 = 12 + 1^2 = 13.$$

Marks (5). Correct transformed expectation, 1; correct variance rule, 1; transformed variance, 1; second-moment identity, 1; final second moment, 1.

Common mistakes. Adding 5 to the variance; multiplying variance by -2 rather than 4; using $\mathbb{E}[Y^2] = \mathrm{Var}(Y) + \mathbb{E}[Y]$.

Solution to `W03-ST04-PA (S3, S4)` Method and conditions. Use the Poisson PMF on the non-negative integers. Under the course parameterisation, the parameter is both the mean and the variance.

The support is $\{0, 1, 2, ...\}$. Thus

$$\begin{aligned}\mathbb{P}(N \leq 1) &= \mathbb{P}(N = 0) + \mathbb{P}(N = 1) \\ &= e^{-3}\frac{3^0}{0!} + e^{-3}\frac{3^1}{1!} \\ &= 4e^{-3}.\end{aligned}$$

Also,

$$\mathbb{E}[N] = 3, \qquad \mathrm{Var}(N) = 3.$$

Marks (5). Correct support, 1; Poisson PMF setup, 1; two probability terms, 1; simplified probability, 1; mean and variance, 1.

Common mistakes. Beginning the support at one; omitting 0!; treating the parameter as a success probability; using different Poisson mean and variance values.

Record the first cause of each error under `M5`, `S3` or `S4`, then reattempt that part from a blank page in two or three days.